

高等微积分 (2)

第十五章: 曲面、曲线、切平面与法线

15.1 曲面、曲线、切平面与法线

复习: 三维空间中的直线与平面

取 $P_0(x_0, y_0, z_0) \in \mathbb{R}^3$. 设 $\vec{e} = (a, b, c)^T \in \mathbb{R}^3$ 为非零向量.

过 P_0 沿方向 $\vec{e}$ 的直线 Γ 的方程为

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

该直线也可以表示成

$$\begin{cases} x = x_0 + at, \\ y = y_0 + bt, \\ z = z_0 + ct. \end{cases}$$

过 P_0 并且与 Γ 垂直的平面 S 称为 Γ 过 P_0 的

法平面, 它的方程为 $\begin{pmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0,$

也就是说 $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$

我们称 $\vec{e}$ 为平面 S 的法向量, Γ 为 S 的法线.

曲面及其切平面和法线

第一种类型：曲面的显函数表示法：

曲面 $S : z = f(x, y)$, $(x, y) \in D \subset \mathbb{R}^2$. 假设 f 在点 (x_0, y_0) 处可微. 令 $z_0 = f(x_0, y_0)$.

当 $(x, y) \rightarrow (x_0, y_0)$ 时,

$$\begin{aligned} f(x, y) - z_0 &= f(x, y) - f(x_0, y_0) \\ &= \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) \\ &\quad + o(\sqrt{(x - x_0)^2 + (y - y_0)^2}). \end{aligned}$$

则曲面 S 在点 (x_0, y_0, z_0) 处的切平面方程为

$$z - z_0 = \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0).$$

即:

$$\frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) + (-1)(z - z_0) = 0.$$

或者

$$\frac{\partial f}{\partial x}(x_0, y_0)x + \frac{\partial f}{\partial y}(x_0, y_0)y + (-1)z = D$$

于是该切平面的法向量为

$$\vec{n} = \begin{pmatrix} \frac{\partial f}{\partial x}(x_0, y_0) \\ \frac{\partial f}{\partial y}(x_0, y_0) \\ -1 \end{pmatrix}.$$

相应的法线方程为

$$\frac{x - x_0}{\frac{\partial f}{\partial x}(x_0, y_0)} = \frac{y - y_0}{\frac{\partial f}{\partial y}(x_0, y_0)} = \frac{z - z_0}{-1}.$$

第二种类型: 曲面的参数表示法:

考虑曲面 S :
$$\begin{cases} x = f_1(u, v), \\ y = f_2(u, v), \\ z = f_3(u, v), \end{cases} \quad (u, v) \in D \subset \mathbb{R}^2.$$

设 $(u_0, v_0) \in D$, f_1, f_2, f_3 在点 (u_0, v_0) 可微. 令

$$\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} = \begin{pmatrix} f_1(u_0, v_0) \\ f_2(u_0, v_0) \\ f_3(u_0, v_0) \end{pmatrix}.$$

当 $(u, v) \rightarrow (u_0, v_0)$ 时, 我们有

$$\begin{pmatrix} f_1(u, v) - x_0 \\ f_2(u, v) - y_0 \\ f_3(u, v) - z_0 \end{pmatrix} = \frac{\partial(f_1, f_2, f_3)}{\partial(u, v)}(u_0, v_0) \begin{pmatrix} u - u_0 \\ v - v_0 \end{pmatrix} + \vec{o}(\sqrt{(u - u_0)^2 + (v - v_0)^2}).$$

当矩阵 $\frac{\partial(f_1, f_2, f_3)}{\partial(u, v)}(u_0, v_0)$ 的秩等于 2 时, 曲面 S 在点 (x_0, y_0, z_0) 处有切平面

$$\begin{pmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{pmatrix} = \frac{\partial(f_1, f_2, f_3)}{\partial(u, v)}(u_0, v_0) \begin{pmatrix} u - u_0 \\ v - v_0 \end{pmatrix} \dots\dots\dots \clubsuit$$

即

$$\begin{pmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \\ \frac{\partial f_3}{\partial u} & \frac{\partial f_3}{\partial v} \end{pmatrix} \begin{pmatrix} u - u_0 \\ v - v_0 \end{pmatrix}$$

展开得

$$x - x_0 = \frac{\partial f_1}{\partial u} \cdot (u - u_0) + \frac{\partial f_1}{\partial v} \cdot (v - v_0);$$

$$y - y_0 = \frac{\partial f_2}{\partial u} \cdot (u - u_0) + \frac{\partial f_2}{\partial v} \cdot (v - v_0);$$

该切平面也可以表示成

$$\begin{aligned} & \frac{D(f_2, f_3)}{D(u, v)}(u_0, v_0)(x - x_0) + \frac{D(f_3, f_1)}{D(u, v)}(u_0, v_0)(y - y_0) \\ & + \frac{D(f_1, f_2)}{D(u, v)}(u_0, v_0)(z - z_0) = 0. \end{aligned}$$

从而曲面 S 在点 (x_0, y_0, z_0) 处的法线方程为

$$\frac{x - x_0}{\frac{D(f_2, f_3)}{D(u, v)}(u_0, v_0)} = \frac{y - y_0}{\frac{D(f_3, f_1)}{D(u, v)}(u_0, v_0)} = \frac{z - z_0}{\frac{D(f_1, f_2)}{D(u, v)}(u_0, v_0)}.$$

第三种类型: 曲面的隐函数表示法: 考虑 $S: F(x, y, z) = 0$.

设 $P_0(x_0, y_0, z_0) \in S$, 而 F 在点 P_0 处可微. 则

当 $S \ni P(x, y, z) \rightarrow P_0$ 时, 我们有

$$\begin{aligned} 0 &= F(x, y, z) - F(x_0, y_0, z_0) \\ &= \frac{\partial F}{\partial x}(P_0)(x - x_0) + \frac{\partial F}{\partial y}(P_0)(y - y_0) \\ &\quad + \frac{\partial F}{\partial z}(P_0)(z - z_0) + o(\|P - P_0\|). \end{aligned}$$

从而当 $J_F(P_0) \neq \vec{0}$ 时, 曲面在点 P_0 有切平面

$$\frac{\partial F}{\partial x}(P_0)(x - x_0) + \frac{\partial F}{\partial y}(P_0)(y - y_0) + \frac{\partial F}{\partial z}(P_0)(z - z_0) = 0.$$

于是曲面 S 在点 P_0 处的法向量为

$$\vec{n} = \begin{pmatrix} \frac{\partial F}{\partial x}(P_0) \\ \frac{\partial F}{\partial y}(P_0) \\ \frac{\partial F}{\partial z}(P_0) \end{pmatrix} = \text{grad} F(P_0),$$

相应的法线方程为 $\frac{x-x_0}{\frac{\partial F}{\partial x}(P_0)} = \frac{y-y_0}{\frac{\partial F}{\partial y}(P_0)} = \frac{z-z_0}{\frac{\partial F}{\partial z}(P_0)}.$

我们也可以从另外的角度来看待上述情形.

设 F 为 $\mathcal{C}^{(1)}$ 类且 $\frac{\partial F}{\partial z}(P_0) \neq 0$. 由隐函数定理知, 局部上我们有 $z = f(x, y)$. 于是在点 P_0 处的切平面方程为

$$\frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) - (z - z_0) = 0.$$

$$-\frac{\frac{\partial F}{\partial x}(P_0)}{\frac{\partial F}{\partial z}(P_0)}(x - x_0) - \frac{\frac{\partial F}{\partial y}(P_0)}{\frac{\partial F}{\partial z}(P_0)}(y - y_0) - (z - z_0) = 0.$$

$$\frac{\partial F}{\partial x}(P_0)(x - x_0) + \frac{\partial F}{\partial y}(P_0)(y - y_0) + \frac{\partial F}{\partial z}(P_0)(z - z_0) = 0.$$

例 1. 设曲面 S 的参数方程为

$$\begin{cases} x = u + e^{u+v}, \\ y = u + v, \\ z = e^{u-v}. \end{cases}$$

求 S 在 $u_0 = 1, v_0 = -1$ 处的切平面与法线.

解: 由题设可知

$$\frac{\partial(x, y, z)}{\partial(u, v)} = \begin{pmatrix} 1 + e^{u+v} & e^{u+v} \\ 1 & 1 \\ e^{u-v} & -e^{u-v} \end{pmatrix}.$$

于是我们有

$$\frac{\partial(x, y, z)}{\partial(u, v)} \Big|_{(1, -1)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ e^2 & -e^2 \end{pmatrix}.$$

由此立刻知矩阵 $\frac{\partial(x, y, z)}{\partial(u, v)} \Big|_{(1, -1)}$ 的秩等于 2, 从而所求切平面的参数方程为(用前面的公式♣)

$$\begin{pmatrix} x - 2 \\ y \\ z - e^2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ e^2 & -e^2 \end{pmatrix} \begin{pmatrix} u - 1 \\ v + 1 \end{pmatrix}.$$

该切平面也可表示成

$$-2e^2(x-2) + 3e^2y + 1(z-e^2) = 0.$$

进而可知所求法线方程为

$$\frac{x-2}{-2e^2} = \frac{y}{3e^2} = \frac{z-e^2}{1}.$$

也可以写成:

$$\frac{x-2}{2e^2} = \frac{y}{-3e^2} = \frac{z-e^2}{-1}.$$

例 2. 若两曲面在交线上每点的法线互相垂直, 则称二者正交. 求证: 曲面 $S_1: F_1(x, y, z) = 0$ 和曲面 $S_2: F_2(x, y, z) = 0$ 正交的充分必要条件是对于交线上的每点 $P_0(x_0, y_0, z_0)$, 均有

$$\frac{\partial F_1}{\partial x} \frac{\partial F_2}{\partial x} + \frac{\partial F_1}{\partial y} \frac{\partial F_2}{\partial y} + \frac{\partial F_1}{\partial z} \frac{\partial F_2}{\partial z} = 0.$$

证明: 在上述两曲面的交点 P_0 , 它们的法向量分别为 $\text{grad}F_1(P_0)$, $\text{grad}F_2(P_0)$, 二者正交当且仅当 $\text{grad}F_1(P_0) \cdot \text{grad}F_2(P_0) = 0$. 由此得证.

例 3. 求证: 球面 $S_1 : x^2 + y^2 + z^2 = R^2$ 与锥面 $S_2 : x^2 + y^2 = a^2 z^2$ 正交.

证明: $\forall (x, y, z) \in \mathbb{R}^3$, 定义

$$F_1(x, y, z) = x^2 + y^2 + z^2 - R^2,$$

$$F_2(x, y, z) = x^2 + y^2 - a^2 z^2.$$

于是 $\forall P(x, y, z) \in S_1 \cap S_2$, 我们有

$$\begin{aligned} & \operatorname{grad} F_1(P) \cdot \operatorname{grad} F_2(P) \\ &= (2x) \cdot (2x) + (2y) \cdot (2y) + (2z) \cdot (-2a^2 z) \\ &= 4(x^2 + y^2 - a^2 z^2) = 4F_2(P) = 0. \end{aligned}$$

故所证结论成立.

空间曲线及其切线和法平面

第1 种类型: 空间曲线的参数表示法:

$$\Gamma : \begin{cases} x = x(t), \\ y = y(t), \\ z = z(t), \end{cases} \quad t \in [\alpha, \beta].$$

若上述函数在点 $t = t_0$ 处可微, 则称曲线 Γ 在相应点 $P_0(x_0, y_0, z_0)$ 处可微, 相应切线方程为

$$\begin{cases} x - x_0 = x'(t_0)(t - t_0), \\ y - y_0 = y'(t_0)(t - t_0), \\ z - z_0 = z'(t_0)(t - t_0). \end{cases}$$

该切线也可表述成

$$\frac{x - x_0}{x'(t_0)} = \frac{y - y_0}{y'(t_0)} = \frac{z - z_0}{z'(t_0)},$$

这里需要假设 $(x'(t_0), y'(t_0), z'(t_0))$ 不为零向量. 我们将经过点 P_0 并且与上述切线垂直的平面称为 Γ 在点 P_0 处的法平面, 其方程为

$$x'(t_0)(x - x_0) + y'(t_0)(y - y_0) + z'(t_0)(z - z_0) = 0.$$

第2 种类型: 空间曲线的隐函数表示法:

$$\Gamma : \begin{cases} F_1(x, y, z) = 0, \\ F_2(x, y, z) = 0. \end{cases}$$

设 F_1, F_2 在点 $P_0(x_0, y_0, z_0)$ 可微且 $\text{grad}F_1(P_0)$, $\text{grad}F_2(P_0)$ 不为零, 则曲线 Γ 在该点的切线为

$$\begin{cases} \frac{\partial F_1}{\partial x}(P_0)(x - x_0) + \frac{\partial F_1}{\partial y}(P_0)(y - y_0) + \frac{\partial F_1}{\partial z}(P_0)(z - z_0) = 0, \\ \dots\dots\dots \clubsuit \\ \frac{\partial F_2}{\partial x}(P_0)(x - x_0) + \frac{\partial F_2}{\partial y}(P_0)(y - y_0) + \frac{\partial F_2}{\partial z}(P_0)(z - z_0) = 0. \\ \dots\dots\dots \spadesuit \end{cases}$$

该切线的方向为

$$\begin{aligned}\vec{T} &= \text{grad}F_1(P_0) \times \text{grad}F_2(P_0) \\ &= \begin{pmatrix} \frac{\partial F_1}{\partial x}(P_0) \\ \frac{\partial F_1}{\partial y}(P_0) \\ \frac{\partial F_1}{\partial z}(P_0) \end{pmatrix} \times \begin{pmatrix} \frac{\partial F_2}{\partial x}(P_0) \\ \frac{\partial F_2}{\partial y}(P_0) \\ \frac{\partial F_2}{\partial z}(P_0) \end{pmatrix} = \begin{pmatrix} \frac{D(F_1, F_2)}{D(y, z)}(P_0) \\ \frac{D(F_1, F_2)}{D(z, x)}(P_0) \\ \frac{D(F_1, F_2)}{D(x, y)}(P_0) \end{pmatrix}.\end{aligned}$$

只有当 $\vec{T} \neq \vec{0}$ 时, 上述方程组才的确给出一条直线. 此时 Jacobi 矩阵 $\frac{\partial(F_1, F_2)}{\partial(x, y, z)}(P_0)$ 的秩等于 2.

借助 $\vec{T}$, 我们也可得到切线的另外一个表述:

$$\frac{x - x_0}{\frac{D(F_1, F_2)}{D(y, z)}(P_0)} = \frac{y - y_0}{\frac{D(F_1, F_2)}{D(z, x)}(P_0)} = \frac{z - z_0}{\frac{D(F_1, F_2)}{D(x, y)}(P_0)}.$$

例 4. 求曲线

$$\begin{cases} F_1(x, y, z) = x^2 + y^2 + z^2 - 9 \\ F_2(x, y, z) = xy - z \end{cases}$$

在点 $P_0(1, 2, 2)$ 处的切线方程与法平面方程.

解: 由题设可知

$$\frac{\partial(F_1, F_2)}{\partial(x, y, z)}(P_0) = \begin{pmatrix} 2x & 2y & 2z \\ y & x & -1 \end{pmatrix} \Big|_{P_0} = \begin{pmatrix} 2 & 4 & 4 \\ 2 & 1 & -1 \end{pmatrix}.$$

于是所求切线的方程为(用♣ + ♠):

$$\begin{cases} 2(x-1) + 4(y-2) + 4(z-2) = 0, \\ 2(x-1) + (y-2) - (z-2) = 0. \end{cases}$$

进而可知所求切线的方向为

$$\vec{T} = \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -8 \\ 10 \\ -6 \end{pmatrix}.$$

于是所求切线的方程为 $\frac{x-1}{-8} = \frac{y-2}{10} = \frac{z-2}{-6}$, 相应法平面方程为

$$-8(x-1) + 10(y-2) - 6(z-2) = 0.$$

同学们辛苦了!